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Due 3/8/26 (extended to 3/11/26)

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1. $T(n)$ is the average-case complexity of badClosestpair

$$T(n) = \sum_{r} \text{Prob}(r) \left(T(r) + T(n-r) + O(r(n-r)) \right)$$

where

- $r = |A_1|$
- $n - r = |A_2|$

The term $O(r(n-r))$ comes from computing all cross-pair distances.

r follows a distribution centered at $n/2$ (approximately normal).
Thus with high probability

$$r \approx \frac{n}{2}$$

So we approximate the recurrence using $r = n/2$.

Guess average-case complexity of badClosestpair

$$T(n) = cn^2 \log n$$

Substitute $r \approx n/2$:

$$T(r) + T(n-r) \approx 2T(n/2)$$

$$= 2c(n/2)^2 \log(n/2)$$

$$= \frac{cn^2}{2} \log(n/2)$$

$$r(n-r)$$

Substitute $r \approx n/2$:

$$r(n-r) \approx (n/2)(n/2)$$

$$= \frac{n^2}{4}$$

$$O(n^2)$$

$$T(n) \approx \frac{cn^2}{2} \log(n/2) + O(n^2)$$

Use

$$\log(n/2) = \log n - 1$$

Then

$$T(n) = \frac{cn^2}{2}(\log n - 1) + O(n^2)$$

$$= \frac{cn^2}{2}\log n - \frac{cn^2}{2} + O(n^2)$$

For sufficiently large c

$$-\frac{cn^2}{2}$$

absorbs the $O(n^2)$ term.

$$T(n) \leq cn^2 \log n$$

$$T(n) = O(n^2 \log n)$$

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2. Compute the worst-case complexity of naiveKarasuba

$$\sum_{k=1}^{n-1} O(kn)^{1.59})$$

$$= O(n^{1.59}) \sum_{k=1}^{n-1} k^{1.59}$$

$$= O(n^{1.59} * n^{2.59})$$

$$= O(n^{4.18})$$

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3. use $BK(m, n)$ to denote worst-case complexity of betterKartsuba running on m numbers of n-bit strings.

- complexity of betterKartasuba over first group: $BK(\frac{n}{2}, 2)$
- complexity of betterKartasuba over second group: $BK(\frac{n}{2}, 2)$

$$\sum_{\mu - r <= r <= \mu + \sigma} cr^2 \log r$$

$$<= c \int_{\mu-\sigma}^{\mu+\sigma} x^2 \log x \; dx$$

$$= c * (\frac{x^3}{3} * \log x \; - \frac{x^3}{9 \ln 2}) \; |_{\mu-\sigma}^{\mu+\sigma}$$

$$<= 2c * \mu^2 * \log \mu * \sigma$$

$$\mu = \frac{n}{2}$$

$$\sigma = \frac{\sqrt{n}}{2}$$

$$= 2c * \frac{n^2}{4} * \log\frac{n}{2} * \frac{\sqrt{n}}{2}$$

$$<= \frac{1}{4} * n^2 \log n * \sqrt{n}$$

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4. There are n^3 airplanes inside a cube of size n.
   Divide space into unit cubes (size 1). Since there are n^3 unit cubes and n^3 airplanes, either:
   Each cube has exactly one airplane, so the closest distance is at most

$$\sqrt{3} < 4$$

Some cube has two airplanes, so the closest distance is also at most

$$\sqrt{3} < 4$$

Thus the closest pair distance is always ≤ 4.
Now consider eight-cubes (size 8). Any pair within distance 4 must lie inside the same eight-cube.
There are

$$O(n^3)$$

eight-cubes.
Each eight-cube contains at most 16^3 half-cubes, and each half-cube holds at most one airplane.
So each eight-cube has constant points.
Finding the closest pair inside one eight-cube takes

$$O(1)$$

$$O(n^3) \times O(1) = O(n^3)$$

$$T(n) = O(n^3)$$

5.

For each string a, compute the point

$$(\#0(a), \#1(a))$$

where $\#0(a)$ is the number of 0s and $\#1(a)$ is the number of 1s.
From n strings we get n points.
Run the closest pair algorithm on these points to find the minimum distance $dist$.
The smallest difference between two strings is

$$dist^2$$

Computing the points takes $O(nm)$.
Closest pair takes $O(n \log n)$.

$$O(nm) + O(n \log n)$$

$$m < \log n$$

then two strings must be identical, so the smallest difference is 0 and we skip closest pair.

$$O(nm)$$